

SDN: Evolution, Future, and Modern Controllers

1 The Evolution of SDN

While SDN seems recent, its roots go back to early research on the separation of data and control planes.

Historical Milestones

- **2004 Proposals:** Feamster and Lakshman argued for planes separation.
- **Ethane Project (2007):** Pioneered flow-based Ethernet switches with a centralized controller. Forwarded unmatched packets to the controller for decisions.
- **Transition:** Ethane quickly evolved into the **OpenFlow** project.

2 The Future of SDN

- **Commoditization:** Replacing monolithic hardware with simple commodity switches and sophisticated software.
- **Network Functions Virtualization (NFV):** Extending SDN concepts to virtualize middleboxes (media caching, firewalls) on standard servers.
- **Inter-AS SDN:** Ongoing research into extending SDN from local networks to inter-domain internet routing.

3 Modern SDN Controllers

3.1 OpenDaylight (ODL)

ODL is a highly flexible, open-source controller architecture.

Service Abstraction Layer (SAL)

The SAL allows different applications and devices to communicate regardless of the underlying protocol.

- **AD-SAL (API-Driven):** Applications communicate with controller modules via REST over HTTP.

- **MD-SAL (Model-Driven):** Uses **YANG** data modeling to define network configuration and state. Uses **NETCONF** for device management.

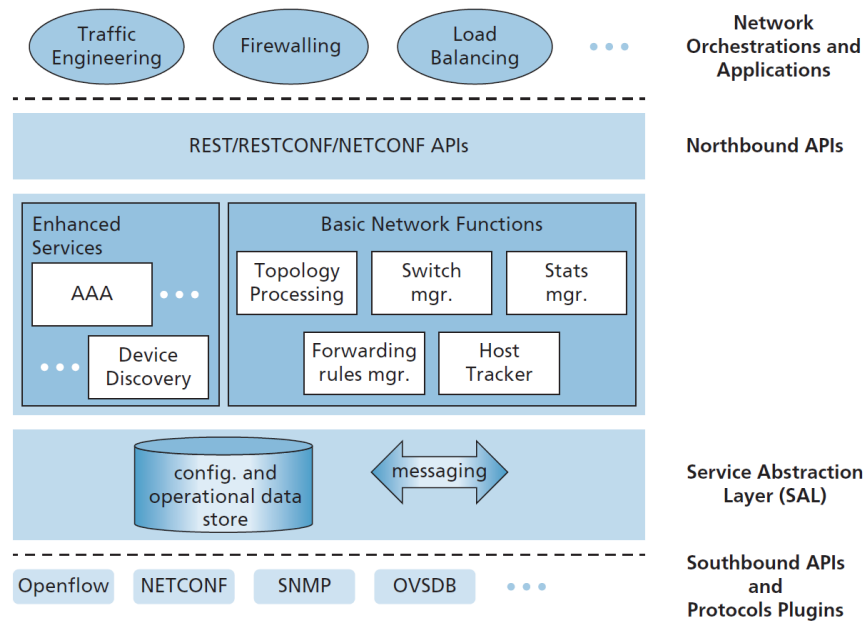


Figure 1: Simplified view of the OpenDaylight controller architecture.

3.2 ONOS (Open Network Operating System)

ONOS focuses on high performance and carrier-grade scalability.

Unique Features of ONOS

- **Intent Framework:** Applications request high-level services (e.g., "Host A cannot talk to Host B") without needing to know low-level configuration details.
- **Distributed Core:** Deployed across a set of interconnected servers, each running identical software, providing fault tolerance and high capacity.

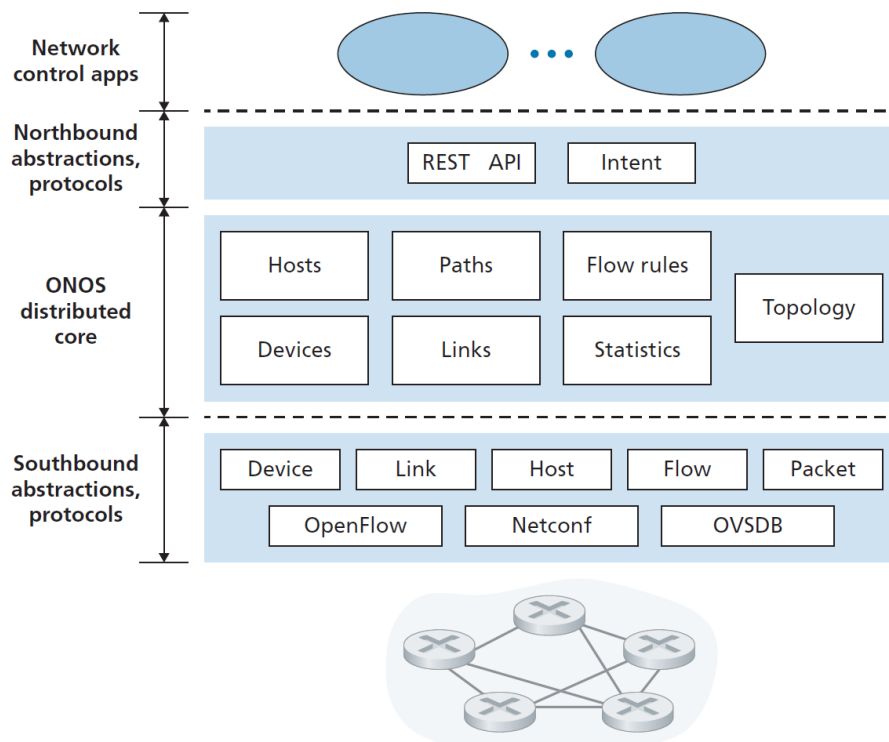


Figure 2: Layered architecture of the ONOS controller.

Summary of Modern Trends

Future architectures move away from proprietary "black boxes" toward open, model-driven, and intent-based systems that prioritize scalability and innovation.